What This System Does

RECLAIM's metals subsystem melts down aluminum scrap recovered from waste streams using a microwave heated cavity at 2.45GHz and 6kW. Shredded metal is loaded into a sodium silicate crucible at the center of a stainless-steel chamber. Microwaves couple into the metal and surrounding susceptor material, raising the crucible to aluminum melting point of 660°C. Molten aluminum is then poured into a sand-cast mold to form mission-relevant hardware.

Key parameters still under evaluation:

- Regolith casting: Lunar regolith as a sand substitute is a candidate but requires binding agent testing before adoption
- Mold Geometries: Mold geometry is adaptable via sand casting; the final part geometries depend on mission requirements and are mot fixed at this stage of the design
- Finalize unknowns (i.e. Lunar Regolith)

Feed Stock

The current feedstock is an aluminum pipe (alloy TBD(?) nominally 6061 or equivalent). Physical characteristics:

Property	Value	Notes
Material	Aluminum (6061 or equivalent(?))	Most common structural Al alloy
Wall thickness	1/10" (0.100")	Per materials spec
Outer diameter	0.5"	Per materials spec
Length	12"	Input stock length; shredded before loading
Melting point	660°C	Pure Al. Alloy may vary ±15°C
Density	2.70 g/cm³	Standard Al density
Pre-process form	Shredded scrap, ~1–2 cm pieces(?)	Maximizes microwave coupling surface area

Materials used

The only material (piece) used in this process will be aluminum pipes with a thickness of 1/10", a diameter of 0.5", and a length of 12". This pipe will need to be shredded down into small scraps so that efficiency and time can be maximized when melting down the metal inside the chamber.

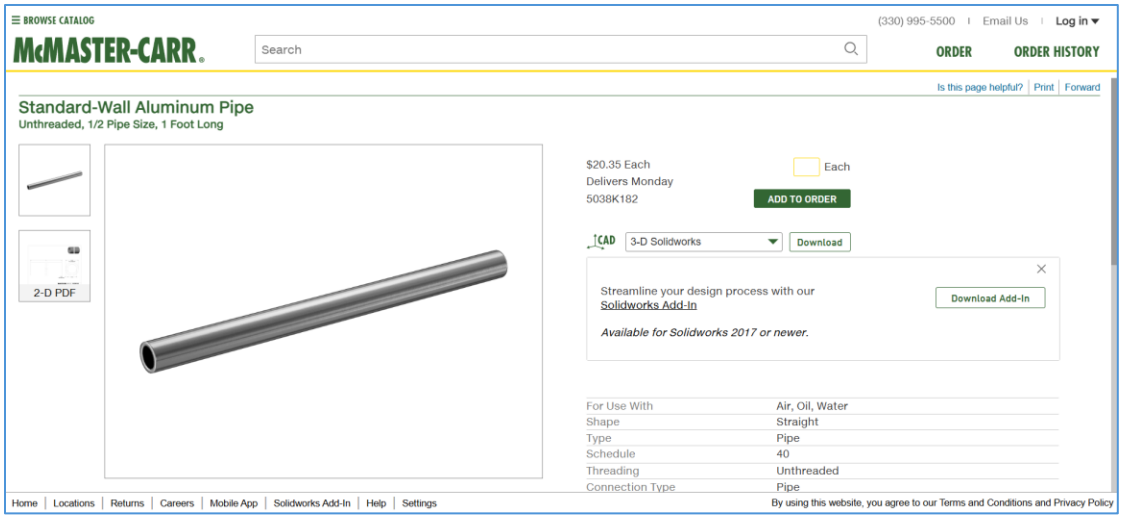


Figure 1 shows the potential feed stock

System Architecture Process Flow

The smelting process follows a linear sequence with no recirculating loops:

FEEDSTOCK (Al pipe stock)

```
    |
    | Mechanical shredding → ~1–2 cm pieces
    ▼
```

CRUCIBLE LOADING (sodium silicate crucible + ceramic fiber insulation blanket)

```
    |
    | Crucible seated at center of SS chamber
    ▼
```

CHAMBER SEAL (bayonet-mount lid engaged)

```
    |
    | 2.45 GHz, 6 kW applied; ramp to 660°C(?)
    ▼
```

MICROWAVE HEATING (melt achieved at 660°C)

```
    |
    | Lid removed; crucible extracted manually by crewmember
    ▼
```

POUR (molten Al poured into sand-cast mold via sprue)

```
    |
    | Air-cool in mold; demold after solidification
    ▼
```

FINISHED PART (cast aluminum component)

Chamber and Lid Design

The smelting chamber is a stainless-steel enclosure chosen for its high-temperature resistance and thermal conductivity. The chamber does not require full vacuum — it operates at ambient or near-ambient pressure, which simplifies lid sealing requirements relative to the pyrolysis subsystem.

Bayonet Mount Lid

Early iterations used clamps and tape for lid retention. This was a failure in testing as the tape melted off at increased temperatures, thus breaking the seal. The current design uses a bayonet mount.

This idea has pegs (a stepped cylindrical peg with a reduced-diameter upper section and a larger diameter bottom portion) on the underside of the lid. The larger diameter base mirrors the radius of holes on the rim of the chamber. These holes are extended into longer slits, matching the reduced-diameter radius. Once the pegs are lined up with the holes, the lid is rotated until the lid can no longer move, and this keeps the lid clamped onto the top of the chamber to contain the heat since the bottom portion of the peg cannot lift. This idea is further supported because the smelt chamber does not need to be a complete vacuum, and the bayonet mount allows the lid to seal the chamber while meeting the necessary specifications.

Parameter	Value	Notes
Chamber material	Stainless steel (grade TBD(?))	High-temp resistance; good microwave shielding
Operating pressure	Ambient (not vacuum)	Simplifies lid seal design
Lid retention	Bayonet mount	Replaces failed tape/clamp approach
Operating temp	660°C target	Aluminum melting point; wall temp lower(?)
Microwave frequency	2.45 GHz	Standard ISM band
Input power	2.2 kW	TBD — may need adjustment for full melt(?)

System Specifications					
Physical Dimensions			**Electrical Requirements**		
System Footprint	Length	160 cm	Microwave System	Input Voltage (AC)	208 V
	Width	53 cm		Max Current Draw	15 A
	Height	76 cm	Conveyance System	Voltage Requirements	12 V
Resonance Chamber	Interior Radius	16 cm		Max Current Draw	5:00 AM
	Interior Height	26.4 cm	**Crucible Dimensions**		
	Waveguide Interface	WR340	Outer Diameter	95mm	
Total System Weight	43 kg (94 lb)		Inner Diameter	85 mm	
			Height	110 mm	
Note: the above specifications are applicable to the current 1.5 kW system. The specifications do not reflect the proposed 6kW final system.					

Crucible

The crucible is made of sodium silicate, and a mold is used to create the shape and size of the crucible we are using. This silicate is perfect for microwave testing since it exhibits resistance to the microwaves at 2.45GHz, which is what the system will be operating at.

The crucible is wrapped in dielectric and ceramic fiber blanket insulation to improve thermal performance, as the crucible temperature otherwise approaches a heat asymptote when left unwrapped in the chamber.

Crucible and Insulation: instructions: How to Make Your Own Crucibles (for microwave metal melting)

Materials:

- Sodium Silicate:
- Sodium Carbide
- Kapton Tape:
- Heavy Duty Aluminum Foil

Crucible Mold: New, Better Crucible Molds for Microwave Metal Melting by ShakeTheFuture | Download STL model | Printables.com

Following the video instructions, we will be able to construct a multitude of crucibles in a variety of shapes, sizes, and intended uses.

Sand Casting

Cast parts are produced via traditional green sandcasting. The method was selected for flexibility (mold geometry is changed by replacing the pattern), low cost, and compatibility with lunar regolith as potential sand substitute in future iterations.

The mold for the molten metal to be poured into is going to be a sand-cast based mold that enables any number of pieces to be made depending on what is needed.

The process of sand casting is first gathering the materials: the cope and drag-- the top and bottom of the box holding the mold, respectively-- the sand, the riser, and the sprue. The riser is made to hold excess molten metal after the mold is filled, and the sprue is the piece where the molten metal flows through.

The drag is placed upside down and filled with wet sand until the sand is flush with the top of the drag. Holes are created in the sand to provide proper ventilation for the fumes. The drag is flipped over, and the pattern, sprue and riser are all placed in the sand, and the cope is placed on top and filled with sand and holes. The box is opened, and the pattern, sprue, and riser are removed. After closing the mold, molten metal is poured through the sprue and then given time for metal to cool. Adding new sand and sifting the used sand can help limit maintenance needed on the system.

This method was chosen through research on feasibility and practicality given the constraints of time and money. Sand casting does not take any more time than other methods but has the tradeoff of needing a crewmate to manually remove the crucible from the chamber and pour the molten aluminum into the mold.

Lunar Regolith as Sand Substitute

Lunar regolith is a soil-like material on the surface of the moon. It is a debris that is formed of dust, rock, and mineral fragments. The debris holds as a promising substitute for sand in the metal-forming process because of its size and temperature tolerance. The particles of the material are very fine, allowing it to create detailed pieces in the mold. Lunar regolith is very refractory, for it can tolerate high temperatures. If the RECLAIM system was used on the Lunar surface, implementation of their material would be efficient because of the abundance of lunar regolith on the Moon.

Although lunar regolith is considered “Moon Soil”, it doesn’t mimic any of the properties of real soil. The debris is inorganic, lacks moisture, and is extremely angular. In order to utilize it in a sand-casting process, it would require moisture. Polymer based binders like Phenolic binders, Epoxy-like resins, and UV-curable polymers can be mixed with the lunar regolith to make it moldable. Other adhesives, like sodium silicate, can also be used as an alkali activator, which transforms the lunar regolith material into something usable.

SMELT System Automation

Current limitations of a manual operation of the smelting process include safety and efficiency concerns. This involves material loading, temperature monitoring, and substantial molten metal handling. To improve reliability, function, and minimize operator interaction, the system will undergo shifts in automation.

The current idea is essentially a locker with an input for the shredded material and an output door to retrieve the ultimately sand-casted piece. Walking through this, after the aluminum is shredded prior to insertion, the user will input the shreds in an upper-level cabinet leading to a funnel. The funnel is fashioned with a bayonet lid and leads to a **graphite/alumina silicate** tube below that funneled into the chamber. The shredded metal travels down this system into the crucible wrapped in insulation and fixed at the bottom of the chamber.

To seal the chamber, a pistol system will be used. Using compressed air, the piston will fire and close the lid on top of the chamber. The rim of the lid will be wrapped in a ceramic gasket to ensure no microwave leakage.

The bottom of the chamber and crucible will have small, aligning holes for the melted aluminum to flow through. As the metal melts, it will drip into the sand cast allowing for a drip cast. As for the sandcast mold, this will be attached externally on a rack-mount system on the bottom of the chamber to align with the aluminum-flow holes. The sandcast mold then can be removed via a drawer at the bottom of the locker. Once the mold is removed, it can be broken to retrieve the

created part. In order to cast again, a new sand mold would have to be created and placed back into the drawer and back onto the mount.